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Fourth Semester, B.Tech Course work
END SEMESTER EXAMINATION, April, 2026

Course code: CAMTC13/COMTC13
Course title: Probability and Stochastic Processes

Time: 3 hours

Maximum Marks. 50

Note: Attempt all questions in given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

<table>
<tr><th>Q. No.</th><th>Questions</th><th>Marks</th><th>CO</th></tr>
<tr><td>Q1</td><td>Attempt any 2 parts of the followings:</td><td></td><td></td></tr>
<tr><td>1a</td><td>A discrete random variable X has the following probability distribution:

X	0	1	2	3	4
$P(X)$	k	$2k$	$3k$	$2k$	k

1. Determine k.
2. Compute $E[X]$, $\text{Var}(X)$ and standard deviation.</td><td>5</td><td>CO1</td></tr>
<tr><td>1b</td><td>A continuous random variable X has the probability density function:
$$f(x) = \begin{cases} k(1-x^2), & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$
1. Determine the value of k.
2. Calculate the first four central moments.
3. Find the coefficient of kurtosis.
4. Comment on the nature of the distribution.</td><td>5</td><td>CO1</td></tr>
<tr><td>1c</td><td>The first four raw moments of a distribution about the origin are given as:
$$\mu_1' = 2, \quad \mu_2' = 10, \quad \mu_3' = 40, \quad \mu_4' = 200$$
1. Obtain the central moments μ_2, μ_3, and μ_4.
2. Verify that $\mu_2 = \mu_2' - (\mu_1')^2$.</td><td>5</td><td>CO1</td></tr>
<tr><td>Q2</td><td>Attempt any 2 parts of the followings:</td><td></td><td></td></tr>
<tr><td>2a</td><td>The following data shows the number of defective items observed in samples of size 4 drawn from a production process:

Number of defectives (x)	0	1	2	3	4
Frequency (f)	32	48	24	8	4

Fit a binomial distribution to the data and find the expected frequencies.</td><td>5</td><td>CO2</td></tr>
<tr><td>2b</td><td>A continuous random variable X follows a Beta distribution with parameters $\alpha > 0$ and $\beta > 0$, having probability density function:
$$f(x) = \begin{cases} \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$
where $B(\alpha, \beta)$ is the Beta function.
1. Derive the mean and variance of the Beta distribution.
2. Find the mean and variance when $\alpha = 3$ and $\beta = 2$.</td><td>5</td><td>CO2</td></tr>
<tr><td>2c</td><td>A random variable X has mean $\mu = 50$ and variance $\sigma^2 = 25$. Using Markov's and Tchebycheff's inequalities, find (i) an upper bound for $P(X \geq 80)$ and (ii) a lower bound for $P(40 \leq X \leq 60)$.</td><td>5</td><td>CO2</td></tr>
</table>

Q. No.	Question	Marks	CO
Q3	Attempt any 2 parts of the followings:		
3a	Find the moment generating function of the Gamma distribution. Hence, obtain the first three cumulants.	5	CO3
3b	State the Central Limit Theorem. Let $\{X_i\}$ be a sequence of independent and identically distributed random variables with mean 3 and variance $\frac{1}{2}$. Let $$S_n = X_1 + X_2 + \cdots + X_n, \quad \text{where } n = 120.$$ Estimate $P(340 \leq S_n \leq 370)$ using the Central Limit Theorem. Given that $P(Z \leq 1.29) = 0.9015$, $P(Z \leq 2) = 0.9772$, and $P(Z \geq 2.58) = 0.0049$.	5	CO3
3c	Fit an exponential curve of the form $Y = ab^x$ to the following data: x: 1, 2, 3, 4, 5, 6, 7, 8 f: 1.0, 1.2, 1.8, 2.5, 3.6, 4.7, 6.6, 9.1	5	CO3
Q4	Attempt any 2 parts of the followings:		
4a	Let p denote the probability that a coin shows a head on a single toss. We wish to test the hypothesis $$H_0 : p = \frac{1}{2} \quad \text{against} \quad H_1 : p = \frac{3}{4}.$$ The coin is tossed 5 times, and H_0 is rejected if more than 3 heads are obtained. Find the probability of a Type I error and the power of the test.	5	CO4
4b	Two researchers adopted different sampling techniques while investigating the same group of students to determine the number of students falling into different intelligence levels. The results are as follows: (see table below) Test whether the sampling techniques adopted by the two researchers differ significantly. Given that $\chi^2_{0.05,2} = 5.991, \chi^2_{0.05,3} = 7.815$.	5	CO4
4c	An insurance agent has claimed that the average age of policyholders who insure through him is less than the average for all agents, which is 30.5 years. A random sample of 100 policyholders who have insured through him gave the following age distribution: (see table below) Calculate the arithmetic mean and standard deviation of the above distribution, and use these values to test the agent's claim at the 5% level of significance. You are given that $Z(1.645) = 0.95$	5	CO4
Q5	Attempt any 2 parts of the followings:		
5a	Define point and interval estimation. Let $x_1, x_2, \ldots, x_n$ be a random sample. Assume that the x_i's are independent Bernoulli random variables of the students picking a course of Statistics with unknown parameter p. Find the maximum likelihood estimator of p, the proportion of students who select the Statistics subject.	5	CO5
5b	The average number of articles produced per day by two machines are 200 and 250, with standard deviations 20 and 25, respectively, based on records of 25 days of production. Can the two machines be regarded as equally efficient at the 1% level of significance? The tabulated value of the t-statistics are $t_{0.005,48} = 2.68, t_{0.01,48} = 2.58$.	5	CO5
5c	Derive the relationship between the t-distribution and the F-distribution.	5	CO5

Table for 3c:

x	1	2	3	4	5	6	7	8
f	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Table for 4b:

Researcher	Below Average	Average	Above Average	Genius	Total
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

Table for 4c:

Age (last birthday)	Number of persons
16–20	12
21–25	22
26–30	20
31–35	30
36–40	16